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Editors
General Relativity and Gravitation

Dear Editors,

Please consider my manuscript, “Surface-Energy Sign Classification for Timelike Kantowski–Sachs–Schwarzschild Junctions on the Ordinary Branch,” for publication in *General Relativity and Gravitation*. The manuscript identifies an invariant causal-sector split. Under the ordinary orientation, every compatible finite-rapidity junction from homogeneous Kantowski–Sachs geometry to the retained Schwarzschild $F > 0$ exterior requires negative Israel surface density, and that ordering persists for a timelike crossing at $F = 0$. In the complementary $F < 0$ sector, with $\epsilon_P = +1$ and the increasing- R Schwarzschild side retained, the ordering becomes sign-indefinite: two distinct future-infalling local embeddings through the same bulk event realize opposite density signs, and an exact compatible choice has $p_s = \sigma/2$, strictly satisfying NEC, WEC, DEC, and the intrinsic shell SEC.

The contribution is the combined causal-sector classification, exact positive-density interval, and explicit regular-chart witness construction, rather than the underlying general BKT junction formula. The Schwarzschild horizon is the invariant boundary between the classified open sectors, while sign indeterminacy begins strictly at $F < 0$. The interior result is local and does not claim a global gluing, dynamics, or validation of generally spacelike transition layers. This combination of exact junction analysis, branch control, and energy-condition analysis is well aligned with the scope of *General Relativity and Gravitation*.

The manuscript is original, is not under consideration elsewhere, and has been prepared independently without institutional or grant support. The manuscript source, figure-generation scripts, and verification suite are publicly available at github.com/RonBibb/junction-ec-paper.

Subject to the journal’s conflict-of-interest checks, potential referees with relevant expertise include João L. Rosa, Sebastian Carloni, S. H. Mazharimousavi, José M. M. Senovilla, and S. Khakshournia.

Sincerely,

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